EXPERIMENT 7: Install Google App Engine and Create a Python Temperature Converter Web Application

1. Aim

To develop and deploy a Python-based temperature converter web application using Google App Engine.

2. Procedure

1. Open Google Cloud Console and sign in with a Google account.
2. Create a new Google Cloud project and select it.
3. Enable billing if required for the project.
4. Install Python and the Google Cloud CLI on the local computer.
5. Open a terminal and initialize the CLI using gcloud init.
6. Sign in and select the Google Cloud project.
7. Create a directory named TemperatureConverter.
8. Create main.py and enter the Flask temperature-conversion application.
9. Create requirements.txt and add Flask as a dependency.
10. Create app.yaml and specify the supported App Engine runtime.
11. Deploy the application using gcloud app deploy.
12. Open the deployed application using gcloud app browse.
13. Enter a Celsius temperature and verify the Fahrenheit result.

3. Program

File – main.py:

```
from flask import Flask, request

app = Flask(__name__)

@app.route("/")
def home():
    c = request.args.get("c")

    if c is None:
        return '''
        <h2>Temperature Converter</h2>
        <form>
          Enter Celsius:
          <input type="number" step="any" name="c" required>
          <input type="submit" value="Convert">
        </form>
        '''

    try:
        celsius = float(c)
        fahrenheit = (celsius * 9 / 5) + 32
        return (f"<h2>Temperature Conversion</h2>"
                f"Celsius: {celsius}<br>"
                f"Fahrenheit: {fahrenheit:.2f}")
    except ValueError:
        return "Enter a valid temperature"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=8080)
```

File – requirements.txt:

```
Flask
```

File – app.yaml:

```
runtime: python313
```

Deployment commands:

```
gcloud init
gcloud app deploy
gcloud app browse
```

4. Output of Program

```
Input: Celsius = 25

Output:
Celsius: 25.0
Fahrenheit: 77.00
```

5. Result

Thus, a temperature converter web application was developed using Python and deployed using Google App Engine.

EXPERIMENT 7: Install Google App Engine and Create a Python Temperature Converter Web Application

1. Aim

To develop a Python-based temperature converter web application using the Flask framework and deploy it on Google App Engine to convert temperature values from Celsius to Fahrenheit.

2. Procedure

1. Open a web browser and navigate to the Google Cloud Console. 2. Sign in using a valid Google account and create a new Google Cloud project. 3. Select the created project and enable the required Google Cloud services. 4. Enable billing if required by the selected App Engine configuration. 5. Install Python on the local computer and verify its installation using python --version. 6. Install the Google Cloud CLI and initialize it using the gcloud init command. 7. Authenticate using the Google account and select the appropriate cloud project. 8. Create a project directory named TemperatureConverter. 9. Inside the directory, create a Python source file named main.py. 10. Import the Flask module and create a Flask application instance. 11. Define a web route that displays an HTML form for accepting the Celsius temperature from the user. 12. Retrieve the entered temperature using the request object provided by Flask. 13. Convert the entered value into a floating-point number to support decimal temperature values. 14. Apply the Celsius-to-Fahrenheit conversion formula: [F=\frac{9}{5}C+32] 1. Display the converted Fahrenheit value on the web page. 2. Add input validation to handle invalid or non-numeric values. 3. Create a requirements.txt file and specify Flask as the required dependency. 4. Create an app.yaml file and specify the supported Python runtime. 5. Open the terminal inside the application directory and execute gcloud app deploy. 6. Wait until the deployment process completes successfully. 7. Execute gcloud app browse to open the deployed web application. 8. Enter a Celsius value such as 25 and verify the corresponding Fahrenheit value.

4. Output of Program

Input: Celsius = 25 Expected Browser Output: Temperature Conversion Celsius: 25.0 Fahrenheit: 77.00 Explanation: The application receives a Celsius value from the user and applies the conversion formula to calculate its equivalent Fahrenheit temperature. For example: [F=\frac{9}{5}\times25+32=77^\circ F] The converted temperature is displayed on the deployed web application.

5. Result

Thus, a temperature converter web application was successfully developed using Python and Flask. The application was deployed on Google App Engine and tested through a browser, successfully converting Celsius values into Fahrenheit.